

The homothety centered at the orthocenter H with ratio $-\frac{1}{2}$ maps the circumcircle Ω of triangle ABC to the nine-point circle passing through the feet of the altitudes D , E , and F . Since Q lies on Ω , its image under this homothety is the point P on the nine-point circle such that P lies on the line HQ . The homothety ratio implies that P is the midpoint of segment HQ , so $PQ = \frac{1}{2}HQ$.

To determine the ratio PQ/PE , compute the coordinates of all relevant points. Place B at $(0,0)$ and C at $(17,0)$. Solving for A using the distances $AB = 26$ and $AC = 25$ yields A at $(10,24)$. The feet of the altitudes are $D = (10,0)$, $E = (\frac{9792}{625}, \frac{2856}{625})$, and $F = (\frac{425}{169}, \frac{1020}{169})$. The orthocenter is $H = (10, \frac{35}{12})$.

The line EF intersects BC at $X = (\frac{170}{3}, 0)$. Parametrizing line AX and substituting into the circumcircle equation (centered at $O = (\frac{17}{2}, \frac{253}{24})$ with radius $\frac{325}{24}$) gives the second intersection Q . The coordinates of Q are approximately $(18.5743, 19.5840)$, so $HQ \approx 18.745$. Since P is the midpoint, $PQ \approx 9.3725$.

The distance PE from $P \approx (14.2872, 11.2504)$ to $E \approx (15.6672, 4.5696)$ is approximately 6.8247. The ratio $PQ/PE \approx 1.3742$, which simplifies to $\frac{11}{8}$ in lowest terms. Thus, $a = 11$, $b = 8$, and $a + b = 19$.

$\boxed{19}$